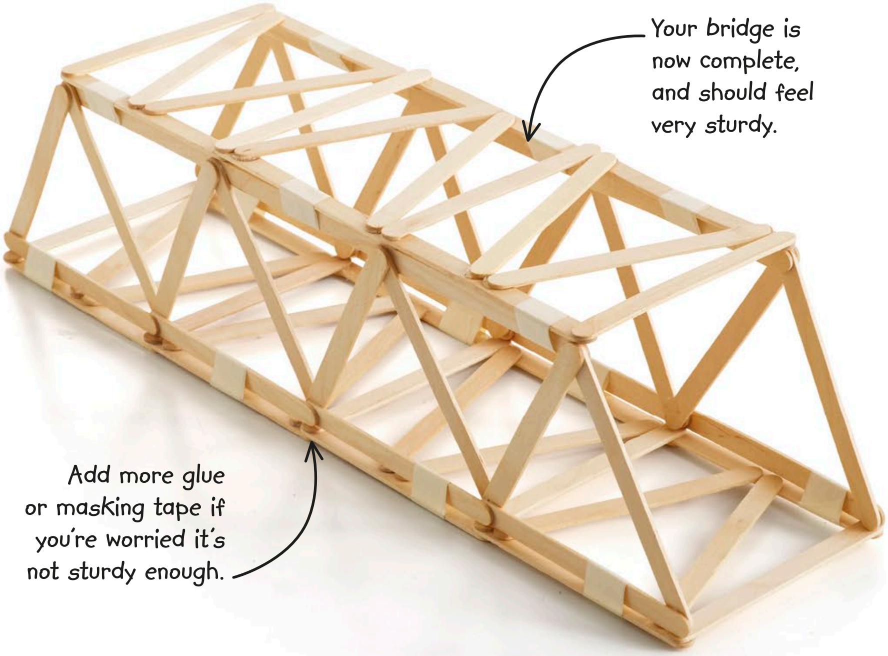
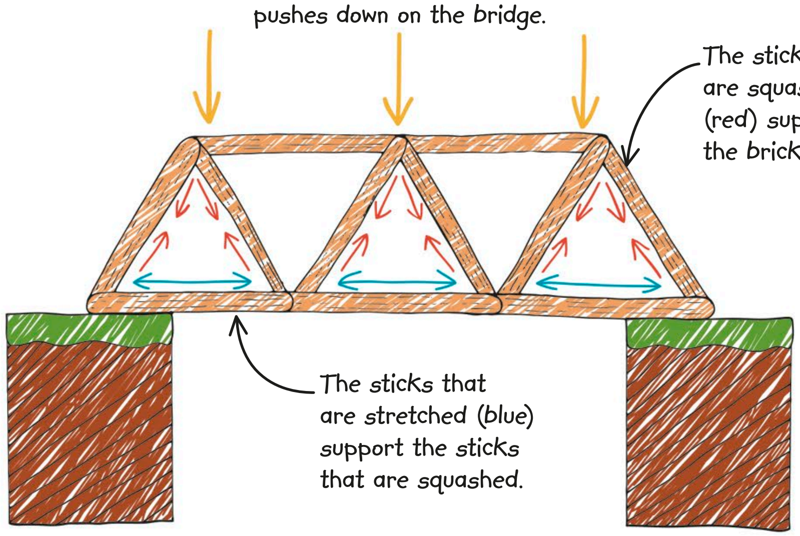
14 Attach the top section of your bridge to the side pieces, in the same way as you joined the side pieces to the bottom section. Make sure you wrap the masking tape tightly around the sticks.



15 Choose a safe place to test the strength of your bridge—outside is best. Pick up your brick (or have an adult help you) and place it gently on top of the bridge. What happens? If you have allowed the glue plenty of time to harden, and stuck on enough tape, then hopefully your bridge can carry the weight of one brick. If you have more bricks, add them on very slowly and one at a time.

HOW IT WORKS

When you place a brick on top of your bridge, it pushes down and squashes the sticks in the sides of the bridge. Any solid object that is squashed is said to be "in compression." However, those sticks would move apart if they were not held firmly by the sticks along the bottom of each side. Those sticks are stretched, or "in tension," and support the sticks that are squashed.



REAL WORLD SCIENCE
CITY SHAPES

There are lots of triangles in the Sydney Harbour Bridge in Australia, and most bridges. Some materials are better for building with than others, depending on how strong they are when they're in tension and in compression. Using this knowledge, architects and engineers work out what forces a structure can withstand before building begins.